

Execution Instructions

DIC2026 Assignment 2

Group05

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1 Project Overview

This document describes how to execute the submitted PySpark implementation for Assignment 2 of the Data-intensive Computing course. The project contains implementations for RDD-based text processing and a Spark DataFrame/ML pipeline.

2 Required Environment

The implementation was tested with the following environment:

- Python 3.11
- Java JDK 17
- PySpark 3.5.1
- NumPy

3 Project Structure

```
Group05_DIC2026_Assignment_2/  
|  
|-- data/  
|   |-- reviews_devset.json  
|   |-- stopwords.txt  
|  
|-- outputs/  
|   |-- output_rdd.txt  
|   |-- output_ds.txt  
|   |-- classification_results.txt  
|  
|-- src/  
|   |-- 00_check_environment.py  
|   |-- 01_part1_rdd_chisquare.py  
|   |-- 02_part2_ds_pipeline.py  
|   |-- utils_preprocessing.py  
|  
|-- requirements.txt  
|-- README.md
```

4 Installing Dependencies

Before running the scripts, install the required Python packages:

```
pip install -r requirements.txt
```

The `requirements.txt` file contains:

```
pyspark==3.5.1  
numpy
```

5 Execution Steps

All scripts should be executed from the `src/` directory.

5.1 Step 1: Check Environment

Run the following command:

```
python 00_check_environment.py
```

This script verifies that PySpark starts correctly and that the development dataset can be loaded.

5.2 Step 2: Run Part 1 - RDD Chi-Square Processing

Run:

```
python 01_part1_rdd_chisquare.py
```

This script performs RDD-based text processing and chi-square term selection.
Generated output:

```
outputs/output_rdd.txt
```

5.3 Step 3: Run Part 2 - Spark DataFrame Pipeline

Run:

```
python 02_part2_ds_pipeline.py
```

This script performs Spark DataFrame-based preprocessing, TF-IDF feature extraction, and classification using Spark ML.

Generated outputs:

```
outputs/output_ds.txt  
outputs/classification_results.txt
```

6 Outputs

The important output files are:

- `output_rdd.txt`: Result of the RDD-based chi-square processing.
- `output_ds.txt`: Result of the Spark DataFrame/ML pipeline.
- `classification_results.txt`: Accuracy and classification result summary.

7 Notes

The implementation uses PySpark only. No scikit-learn or pandas-based processing is used for the main assignment logic. The scripts are designed to run on the development dataset provided for the assignment.